

DSBDAL Assignment 12

MapReduce Log Processing – Complete Command Sequence

1. Create Input Log File (Local)

```
nano log.txt
```

Example content:

```
2026-04-21 10:15:32 INFO User logged in
2026-04-21 10:16:10 ERROR Database connection failed
2026-04-21 10:17:45 WARNING Disk space low
2026-04-21 10:18:20 INFO File uploaded
2026-04-21 10:19:00 ERROR Timeout occurred
```

2. Create Input Directory in HDFS

```
hdfs dfs -mkdir /input
```

3. Upload File to HDFS

```
hdfs dfs -put log.txt /input/
```

Verify:

```
hdfs dfs -ls /input
```

4. Compile Java Files

```
javac -classpath `hadoop classpath` -d . LogMapper.java LogReducer.java
LogDriver.java
```

5. Create JAR File

```
jar -cvf LogAnalysis.jar *
```

Verify JAR exists:

```
ls
```

6. Run MapReduce Job

```
hadoop jar LogAnalysis.jar LogDriver /input/log.txt /output/
```

Common Errors & Fixes

Error 1: JAR file not found

```
JAR does not exist or is not a normal file
```

✓ Fix:

```
ls  
# or  
find . -name "*.jar"
```

Error 2: Output directory already exists

```
FileAlreadyExistsException
```

✓ Fix:

```
hdfs dfs -rm -r /output
```

OR use new directory:

```
hadoop jar LogAnalysis.jar LogDriver /input/log.txt /output1/
```

✗ Error 3: Input path does not exist

```
Input path does not exist
```

✓ Fix:

```
hdfs dfs -ls /input
```

✗ Error 4: Permission denied

✓ Fix:

```
hdfs dfs -chmod -R 777 /input
```

7. View Output

```
hdfs dfs -cat /output/part-r-00000
```

8. Download Output to Local (Optional)

```
hdfs dfs -get /output ./output
```

9. Clean Up (Optional)

```
hdfs dfs -rm -r /input  
hdfs dfs -rm -r /output
```

✓ Expected Output Format

```
INFO      2  
ERROR     2  
WARNING   1
```

Conclusion

The MapReduce program successfully processes the log file and counts occurrences of each log level using Mapper and Reducer phases.